

DATA ANALYSIS INTERNSHIP

Day 5 Report

Student Performance Dataset Expansion & Visualization (Home Assignment)

On Day 5, the **Student Performance Dataset** was further analyzed and organized by categorizing important columns for analysis and prediction. Data visualization techniques were also implemented to represent student performance patterns graphically using different chart types.

1. Expansion of the Dataset (Continuation of Day 2)

The **Student Performance Dataset** was further explored by separating columns based on their importance and relevance for **analysis, prediction, and visualization** purposes.

Available Columns in the Dataset

- student_id
- hours_studied
- sleep_hours
- attendance_percent
- previous_scores
- exam_score
- result (Pass/Fail)
- result_encoded

2. Column Separation

Must Needed Columns

These columns were considered essential for analysis and prediction because they directly influence student performance.

- hours_studied
- previous_scores
- exam_score
- attendance_percent
- sleep_hours

Observation:

These features directly affect **student academic performance and final results**.

Optional Columns

These columns provide additional insights but are not mandatory for prediction.

- student_id
- result_encoded

Observation:

Optional columns help in organizing records and supporting machine learning models.

Target Columns

The following columns were used as target variables for performance prediction.

- exam_score
- result (**Pass/Fail**)

Observation:

The **exam score** and **result** columns determine overall student performance outcomes.

3. Data Visualization of Student Performance

Different visualization techniques were used to represent student performance in a graphical format for easier understanding and analysis.

Visualization helped identify **patterns, comparisons, and relationships** between study behavior and exam outcomes.

4. Charts Created

1. Pie Chart Creation

Objective:

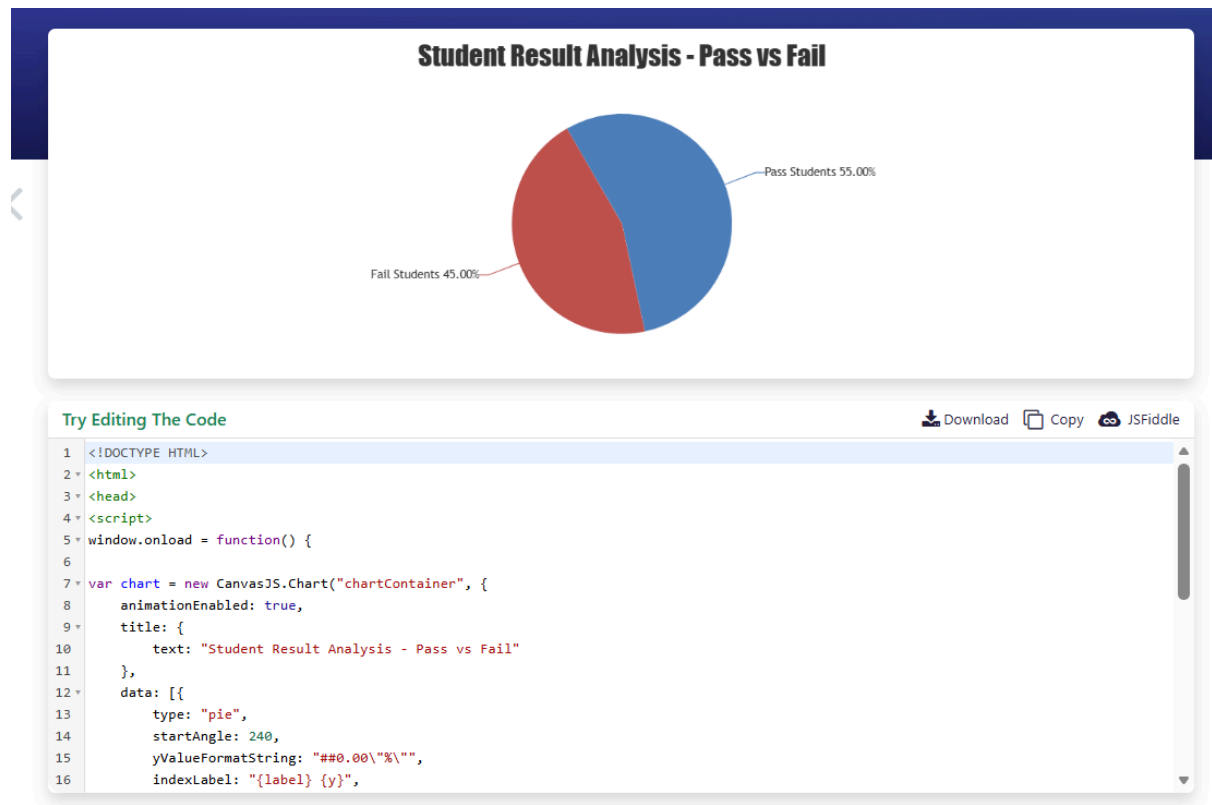
To represent the percentage distribution of student results.

Visualization Used For:

- Pass percentage
- Fail percentage

Observation:

The dataset showed a slightly higher percentage of **Fail students** compared to **Pass students**.



2. Bar Chart Creation

Objective:

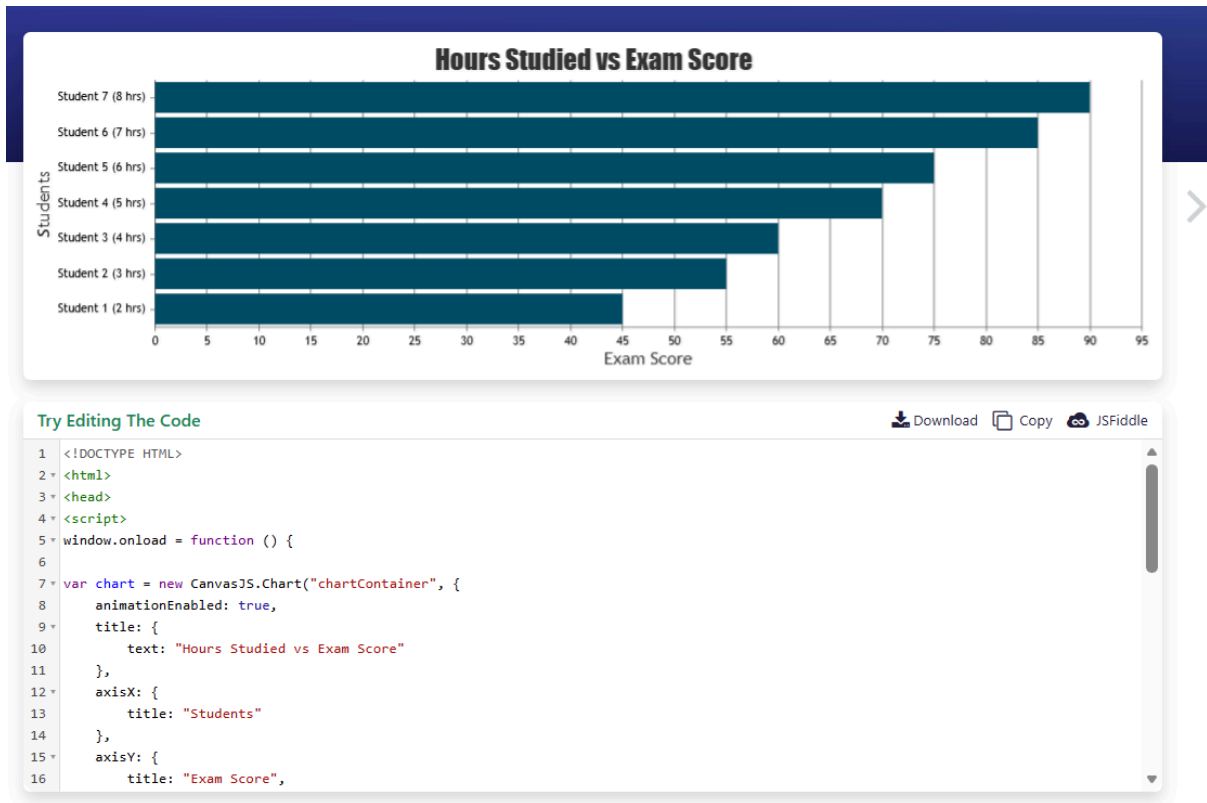
To compare **exam scores** based on **hours studied**.

Visualization Used For:

- Hours studied vs exam score
- Student performance comparison

Observation:

Students with **higher study hours** generally achieved **better exam scores**.



3. Line Chart Creation

Objective:

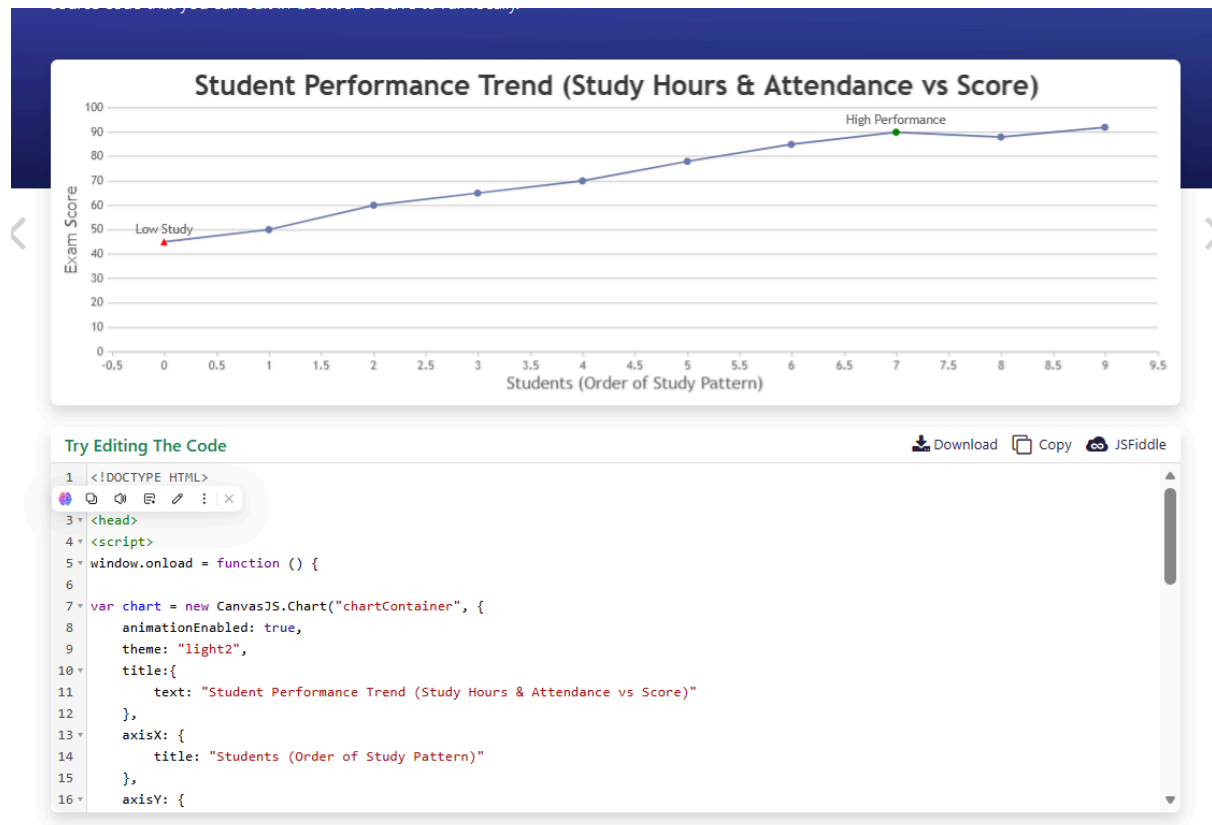
To represent variation in **exam scores** based on **attendance percentage** and **study hours**.

Visualization Used For:

- Study trends
- Score improvement patterns

Observation:

Students with consistent study habits showed better academic performance.



4. Histogram Creation

Objective:

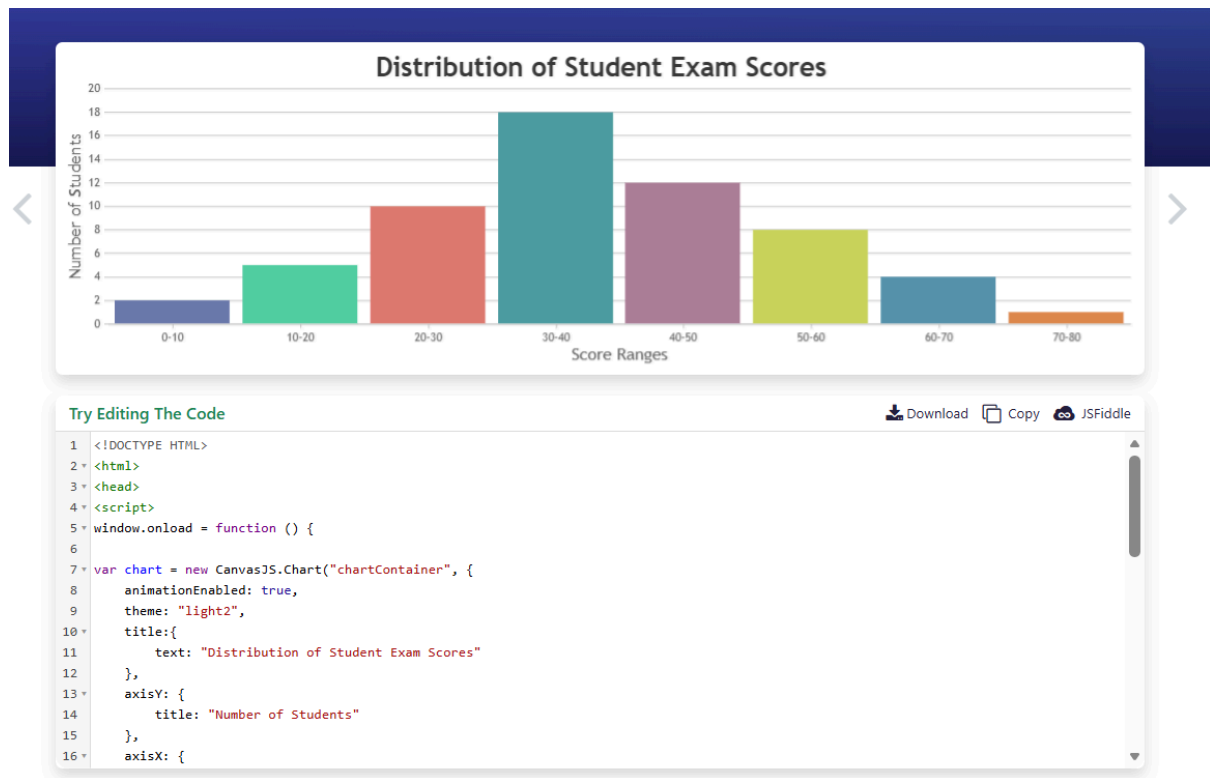
To analyze the distribution of student exam scores.

Visualization Used For:

- Score frequency distribution
- Performance grouping

Observation:

Most student scores were concentrated between **30–40 marks**.



5. Tools Used for Analysis

Google Colab

Google Colab was used to write and execute **Python programs** for data analysis, preprocessing, and visualization tasks.

Features of Google Colab:

- Online Python execution
- Dataset analysis
- Data visualization
- Machine learning support

CanvasJS

CanvasJS was used for creating **interactive and responsive charts** to visually represent student performance data.

Purpose of CanvasJS:

- Create graphical visualizations
 - Compare student performance
 - Represent trends and distributions clearly
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6. Installing Required Modules

Required Python modules were installed using the **pip install** command for performing data analysis and visualization operations.

Common Modules Used:

- pandas
- numpy
- matplotlib
- plotly

Observation:

Installing required libraries helped perform **EDA, preprocessing, and chart visualization efficiently**.

7. Key Insights

Important insights identified from the dataset:

- **Hours studied** strongly influenced exam performance.
 - **Exam score** played the biggest role in deciding **Pass/Fail results**.
 - **Attendance percentage** showed moderate influence on performance.
 - **Sleep hours** had a weaker impact compared to study hours.
 - Graphical visualization made performance comparison easier.
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8. Conclusion

On Day 5, I learned how to **expand and organize datasets** by separating important columns for analysis and prediction. I also created different **visualization charts** to analyze student performance patterns effectively. Tools like **Google Colab** and **CanvasJS** helped in understanding dataset trends, comparisons, and performance insights more clearly.